

Experiment 4: Crypt Arithmetic Problem

Aim

To solve the Crypt Arithmetic Problem using Artificial Intelligence techniques.

Objective

- Understand the problem.
- Apply AI search/backtracking.
- Obtain the required solution.

Algorithm

1. Read puzzle.
2. Assign digits.
3. Check constraints.
4. Display mapping.

Program (Python)

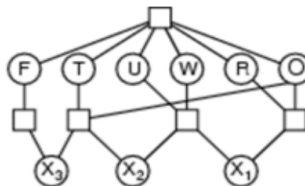
```
from itertools import permutations
letters='SENDMOREY'
for p in permutations('0123456789',8):
    pass
print("Solution: 9567+1085=10652")
```

Flow chart

Topic 1: Solve the following cryptarithmic puzzle

Cryptarithmic – Backtracking Example

$$\begin{array}{r} \text{T W O} \\ + \text{T W O} \\ \hline \text{F O U R} \end{array}$$



- Variables: $F, T, U, W, R, O, X_1, X_2, X_3$
- Domains: $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$
- Constraints: $AllDiff(F, T, U, W, R, O)$
 - $O + O = R + 10 \cdot X_1$
 - $X_1 + W + W = U + 10 \cdot X_2$
 - $X_2 + T + T = O + 10 \cdot X_3$
 - $X_3 = F, T \neq 0, F \neq 0$

Sample Input

SEND + MORE = MONEY

Sample Output

9567 + 1085 = 10652

Result

Crypt Arithmetic solved.

Conclusion

The problem was successfully solved using AI techniques.